Principia Orthogona, Volume I
The Mathematics of Generative Transitions

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Abstract

Every threshold-crossing that cannot be undone — a cell beginning autophagy, a star igniting helium, an elastic rod buckling — follows the same four-step geometric sequence: compression toward a critical surface, curvature intensification, a Whitney fold that breaks injectivity, and stabilisation on a new branch. This paper formalises that sequence as the operator chain $G = U \circ F \circ K \circ C$ acting on trajectories in a Riemannian manifold, derives a free-discontinuity variational principle and symplectic Hamiltonian structure, and proves five structural theorems (existence, non-commutativity, irreducibility, Whitney-fold classification, symplectic preservation), each established by seven independent arguments. Fifty-eight arithmetic and structural facts underlying the framework — including the three canonical constants, Gronwall stability radius $\varepsilon_0 = 1/3$, transverse Lyapunov exponent $\mu_{\max} = -2$, and embodiment threshold $\tau = 2$ — are machine-checked in Lean 4 (AXLE, [9]) against a pinned Mathlib4 revision, with zero `sorry` and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. §17 states precisely which claims that covers and which it does not; the check is reproducible in one command. Theorems A–D (existence, closure, contact normal form, structural stability) are formalised as type-theoretic structures over an arbitrary `MetricSpace` carrier; toy-model constants P1–P6 are separately machine-checked as arithmetic facts. A structural analogy between the five-operator chain and Perelman's Ricci flow with surgery is noted as Conjecture 15.1 (not proved here).

Changes in V7 (August 2026). The mathematics is unchanged. `PrincipiaVol1.lean` is now built and kernel-checked against a pinned Mathlib revision — 58 theorems, zero `sorry` — and reproducible in one command; §17 scopes the claim precisely. The separation theorem is restated in the form its argument supports and proved. One open obligation is added, O7, on the instantiation of ε_0 in §22. Changes in V4–V6 are carried forward unaffected; open obligations O1–O5 remain open. Full account in `CHANGES_Vol1.md`.

Keywords: generative transitions; contact geometry; operator sequence; Whitney fold; singularity theory; variational mechanics; symplectic geometry; Ricci flow; Perelman conjecture; Lean 4 formal verification; dimensional threshold

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1 Introduction

A generative transition — in singularity theory, a *fold singularity of a smooth map* [13] — is what happens when a system crosses a threshold it cannot uncross. The cell that begins to digest itself under nutrient stress. The star that ignites helium when its hydrogen is exhausted. The elastic rod that buckles under axial load. The 3-manifold that develops a geometric singularity under Ricci flow. In each case, a compression drives the system toward a critical point, a curvature instability makes the approach inevitable, a fold commits the system irreversibly to a new branch, and an unfolding establishes the new stable state.

This paper formalises that sequence as four operators acting on trajectories in a Riemannian manifold:

$$C \longrightarrow K \longrightarrow F \longrightarrow U.$$

The second edition adds a fifth operator E (Entropy / Generative Time Circuit), whose action variable accumulates the irreversible cost:

$$C \longrightarrow K \longrightarrow F \longrightarrow U \longrightarrow E \longrightarrow C' \longrightarrow \cdots$$

The central mathematical claim is that this operator sequence is not an analogy or a metaphor: it is a precise structure that admits constructive definitions, analytical invariants, a variational principle, and a Hamiltonian formulation.

The framework is placed within established mathematical traditions: comparison geometry, singularity theory, geometric flows, variational mechanics, and impulsive Hamiltonian systems.

The key insight is that fold singularities appear in every domain where thresholds are crossed irreversibly, and they always arise through the same four-operator geometric sequence. Sections 2–5 make this precise and prove five structural theorems, each by seven independent arguments (§§18–22). Sections 6–12 give computable content. Sections 13–16 connect to dm^3, Perelman, and the dimensional threshold.

What is new. (i) The four-operator decomposition as an axiom system (six assumptions, five theorems); (ii) classification restricted to A_1–A_3 by dimensional count from the assumptions; (iii) free-discontinuity variational principle with jump condition at the fold; (iv) machine-checked constants $(\varepsilon_0, \mu_{\max}, \tau)$ in Lean 4; (v) the seven-proof format: each structural theorem established by independent algebraic, geometric, mechanical, biological, spectral, contact-theoretic, and formal routes.

Related work. The Whitney A_k classification [13, 14] and its singularity-theoretic development [7, 17] underpin the fold operator and Classification Theorem 9.1. Contact geometry follows [6, 18]; symplectic topology [19]. Companion volumes: dm^3 toy model [12]; contact realisation [20]; Volume II [21]. The present paper differs from all of the above in its axiomatic operator-chain treatment and the seven-proof template.

2 Minimal Mathematical Assumptions

Notation. (X, g) Riemannian manifold; $\gamma : [0, T] \to X$ trajectory. $\kappa(s)$ geodesic curvature; $\kappa^*(x)$ fold threshold. $\rho = r - 1$ transverse displacement from limit cycle Γ. $\varepsilon_0 = 1/3$ Gronwall stability radius; $\mu_{\max} = -2$ transverse Lyapunov exponent; $\tau = 2$ embodiment threshold (dm^3 canonical model; see §13). $T^* = 2\pi$ limit-cycle period; $W = \frac{1}{2}\rho^2$ Lyapunov energy. Asymptotics hold as $z \to \infty$ unless noted.

The operator sequence is defined under the weakest assumptions necessary for the constructions of Section 3 to be well-posed.

Assumption 2.1 (Manifold Structure)**.** *The state space X is a smooth, finite-dimensional Riemannian manifold, locally compact and second-countable.*

Assumption 2.2 (Trajectory Regularity). *A system trajectory $\gamma : [0,T] \to X$ is piecewise C^2, locally non-degenerate ($\|\dot{\gamma}(t)\| \neq 0$ a.e.), and has bounded curvature on compact intervals prior to folding events.*

Assumption 2.3 (Compression Feasibility). *There exists a lower-dimensional submanifold $X_C \subset X$ and a Lipschitz projection $C : X \to X_C$ satisfying the non-collapse condition*

$$d\big(C(x_1), C(x_2)\big) \;\geq\; \delta\, d(x_1, x_2)$$

for some $\delta > 0$ and all x_1, x_2 in a compact neighbourhood. This is a bi-Lipschitz embedding condition ensuring distinct trajectories remain distinguishable after compression.

Lemma 2.1 (Bi-Lipschitz alone does not give finitely many pieces). *There exists a bi-Lipschitz map $f : [0,1] \to \mathbb{R}$, injective, satisfying the non-collapse condition of Assumption 2.3 with explicit constant $\delta \approx 0.94$, that is non-differentiable at infinitely many points accumulating at $x = 0$. Hence Assumption 2.3 as stated does not by itself imply C is piecewise smooth with finitely many pieces.*

Proof. Construct $f(x) = x + \sum_{n=1}^{\infty} h_n\, \phi\Big(\frac{x-1/n}{w_n}\Big)$, where $\phi(t) = \max(0, 1 - |t|)$ is the unit tent function and $w_n = 1/(4n^2)$, $h_n = A/n^4$, $A = 3/(20\pi^2)$. The bumps have disjoint supports for every $n \geq 1$ (the gap $1/n - 1/(n+1) = 1/(n(n+1))$ exceeds $w_n + w_{n+1}$ for all $n \geq 1$, by direct algebra), so $f'(x) - 1$ is bounded by $\sup_x |f'(x) - 1| \leq 4A \sum_n n^{-2} = 2A\pi^2/3 = 1/10$, giving f bi-Lipschitz with constants in $[0.9, 1.1]$, while each bump contributes a genuine slope discontinuity at $x = 1/n$, giving a countably infinite singular set accumulating at 0. $\square$ $\square$

Assumption 2.4 (Compression Regularity — new). *The projection C of Assumption 2.3 is, in addition, piecewise C^1 with finitely many pieces on every compact subset of X.*

Assumption 2.4 closes the gap identified in Lemma 2.1 and is assumed throughout the remainder of this paper wherever piecewise smoothness of C is used.

Assumption 2.5 (Curvature Threshold). *The critical curvature is defined intrinsically by the focal radius:*

$$\kappa^*(x) = \frac{1}{\operatorname{foc}(x)},$$

where $\operatorname{foc}(x)$ *is the distance from* x *to the first focal point along the normal direction. In the presence of positive sectional curvature* $K_{\mathrm{sec}}(x) > 0$*, this is modified by the Rauch comparison theorem to*

$$\kappa^*(x) = \min\bigl(\|\mathrm{II}_x\|, \sqrt{K_{\mathrm{sec}}(x)}\bigr),$$

where $\|\mathrm{II}_x\|$ *is the norm of the second fundamental form. For* $K_{\mathrm{sec}}(x) \leq 0$*,* $\kappa^*(x) = \|\mathrm{II}_x\|$*.*

Assumption 2.6 (Folding Well-Posedness)**.** *The folding operator* $F : X_C \to X_F$ *satisfies: the Jacobian* dF *loses rank by exactly 1 at fold points; the fold is local; and the fold produces a finite number of branches.*

Assumption 2.7 (Morse Stability Functional)**.** *The stability functional* $\Phi : X \to \mathbb{R}$ *is* C^2*, bounded below on compact subsets, and Morse: all critical points are non-degenerate, i.e.,* $\nabla^2\Phi(x^*) \succ 0$ *at every local minimum* x^**.*

3 Operator Definitions

Terminology. The *Compression operator* C is a Lipschitz projection onto a lower-dimensional submanifold (standard: nearest-point projection). The *Curvature operator* K is a curvature-driven flow (standard: restricted mean curvature flow [17]). The *Fold operator* F is a smooth map with corank-1 Jacobian loss (standard: Whitney fold [13]). The *Unfold operator* U is gradient flow to a Morse minimum (standard: gradient descent on a Morse function). The *embodiment threshold* τ is the stochastic Lyapunov radius of the limit cycle (standard: Lyapunov stability radius).

Let X be as in Assumption 2.1. A system trajectory is a piecewise C^2 map $\gamma : [0,T] \to X$. The goal is to describe local transitions in γ via the sequence $C \to K \to F \to U$.

Definition 3.1 (Compression Operator C)**.** *A* compression operator *is a map* $C : X \to X_C$ *with* $\dim(X_C) < \dim(X)$ *satisfying Assumption 2.3.* C *reduces degrees of freedom while retaining essential local structure.*

Definition 3.2 (Curvature Operator K)**.** *Given a compressed trajectory* $\gamma_C : [0,T] \to X_C$*, the* curvature operator $K : X_C \to X_C$ *modifies the tangent field by*

$$\frac{d}{ds}\bigl[K(\gamma_C)(s)\bigr] = \dot\gamma_C(s) + \alpha(s)\,\mathbf{n}(s),$$

where $\mathbf{n}(s)$ *is the unit normal and* $\alpha(s) = \lambda\big(\kappa^*(\gamma_C(s)) - \kappa(s)\big)_+$, $\lambda > 0$. *K is a curvature-inducing flow that drives curvature monotonically toward* κ^* *but never beyond it; the sequence is a hybrid dynamical system with a switching condition at* κ^*.

Definition 3.3 (Folding Operator F)**.** *Let* $\gamma_K = K(\gamma_C)$. *A fold occurs at* s_0 *when* $|\kappa_K(s_0)| = \kappa^*(\gamma_K(s_0))$. *The* folding operator $F : X_C \to X_F$ *acts by*

$$F(\gamma_K(s)) = \gamma_K(s) - \beta(s)\,\mathbf{n}(s), \qquad \beta(s) = \begin{cases} 0 & |\kappa_K(s)| < \kappa^*, \\ \mu\big(|\kappa_K(s)| - \kappa^*\big) & |\kappa_K(s)| \geq \kappa^*, \end{cases}$$

where $\mu > 0$ *controls fold depth. At a fold point* s_0, $\mathrm{rank}(dF_{\gamma_K(s_0)}) = \dim(X_C) - 1$.

Definition 3.4 (Unfolding Operator U)**.** *The* unfolding operator $U : X_F \to X$ *is*

$$U(x_F) = \operatorname*{arg\,min}_{y \in \mathcal{N}(x_F)} \Phi(y),$$

where $\mathcal{N}(x_F)$ *is a sufficiently small neighbourhood of* x_F *and* Φ *satisfies Assumption 2.7. The unfolding is realised by the gradient flow* $\dot{y} = -\nabla\Phi(y)$, $y(0) = x_F$, *converging to a non-degenerate local minimum* x^*.

Theorem 3.1 (Sequential Consistency)**.** *Under Assumption 3.1 below, if K is applied until curvature reaches* κ^*, *then F is well-defined, produces a finite branch set, and induces a rank-deficient Jacobian at fold points.*

Lemma 3.2 (Constant κ^* prevents finite-time folding)**.** *If* $\kappa^*(\gamma_C(s)) \equiv k_0$ *is constant along the trajectory near* s_0, *and* $\kappa(\gamma_K(s_0)) < k_0$, *then under the K-dynamics of Definition 3.2,* $\kappa(\gamma_K(s)) < k_0$ *for all finite* $s > s_0$*: the fold event* $|\kappa_K(s)| = \kappa^*$ *does not occur in finite arc-length.*

Proof. With κ^* locally constant at k_0, the ODE of Definition 3.2 becomes $\dot{\kappa} = \lambda(k_0 - \kappa)$ for $\kappa < k_0$ (linear, autonomous), with explicit solution $\kappa(s) = k_0 - Ce^{-\lambda(s-s_0)}$, $C = k_0 - \kappa(\gamma_K(s_0)) > 0$. Since $e^{-\lambda(s-s_0)} > 0$ for every finite s, $\kappa(s) < k_0$ strictly for all finite s: convergence to k_0 is only asymptotic ($s \to \infty$). □ □

Assumption 3.1 (Transverse Crossing — new)**.** *Along the trajectory* γ_C, *whenever* $\kappa(\gamma_K(s))$ *approaches* $\kappa^*(\gamma_K(s))$ *from below, the crossing is transverse:* $\frac{d}{ds}\big(\kappa^*(\gamma_K(s)) - \kappa(\gamma_K(s))\big) \neq 0$ *at the (assumed to exist) crossing point* s_0.

Proof of Theorem 3.1. **Curvature reaches κ^* (finite time, under Assumption 3.1).** By Definition 3.2, $\dot{\kappa} = \alpha(s) = \lambda(\kappa^*(\gamma_C(s)) - \kappa(s))_+ \geq 0$, so κ is non-decreasing and strictly increasing while $\kappa < \kappa^*$. Assumption 3.1 guarantees the crossing occurs at a finite s_0 (Lemma 3.2 shows this fails without it) and is simple (non-degenerate).

Rank-deficient Jacobian at fold points. By Definition 3.3, at s_0, F acts as $F(\gamma_K(s)) = \gamma_K(s) - \beta(s)\mathbf{n}(s)$ with β continuous but only piecewise differentiable at $|\kappa_K| = \kappa^*$: $\beta = 0$ for $\kappa_K < \kappa^*$, $\beta = \mu(|\kappa_K| - \kappa^*)$ for $\kappa_K \geq \kappa^*$, with one-sided derivatives 0 and μ disagreeing at s_0. This is a rank drop from $\dim(X_C)$ to $\dim(X_C)-1$ confined to the normal direction $\mathbf{n}(s_0)$, derived directly from the case-split structure of β.

Finitely many branches. By the Morse condition on Φ, non-degenerate critical points are isolated (a standard consequence of the Morse Lemma via the inverse function theorem applied to $\nabla\Phi$, whose Jacobian at a critical point is $\nabla^2\Phi(x^*)$, invertible by non-degeneracy). On the compact neighbourhood $\mathcal{N}(x_F)$, isolation plus compactness gives finitely many critical points, hence finitely many candidate branches; shrinking $\mathcal{N}(x_F)$ below the minimum pairwise distance between them ensures at most one critical point lies in $\mathcal{N}(x_F)$, so $U(x_F)$ is unique with no tie possible, converging by the stable manifold theorem for gradient flows at a non-degenerate critical point [15, 16]. □ □

4 Falsifiability Conditions

The framework is falsifiable at four levels.

F1 — Compression. If empirical data show expansion under C, or collapse of distinct trajectories, the model fails.

F2 — Curvature. If a fold occurs at curvature strictly below κ^*, or curvature exceeds κ^* without folding, the model is falsified. (κ^* is computed via the focal radius, not post-hoc from fold observations. Operationally: in the Euler–Bernoulli rod, $\kappa^* = \pi^2 EI/L^2$ (standard buckling test); in autophagy, $\kappa^* = \mathrm{IC}_{50}(\mathrm{mTORC1})/\mathrm{foc}(x_{\mathrm{auto}})$ (kinase assay); in the dm^3 toy model, $\kappa^* = 1$ exactly (Lean: P1a–P1d).)

F3 — Folding. If empirical fold events do not correspond to Jacobian rank loss, or produce infinitely many branches, the model fails.

F4 — Sequence order. If transitions occur in a different order, or stabilisation occurs without folding, the model is falsified.

5 Structural Theorems

Theorem 5.1 (Existence and Well-Posedness)**.** *Under Assumptions 2.1–2.7 and 3.1, the composite operator $G = U \circ F \circ K \circ C$ is well-defined on any piecewise C^2 trajectory.*

Proof. We show each stage is single-valued with output in the domain of the next.

C **is well-defined.** Assumption 2.3 gives C as a Lipschitz (hence single-valued, continuous) projection $X \to X_C$, defined on all of X.

K **is well-defined.** By Definition 3.2, K solves the ODE $\frac{d}{ds}\gamma_K(s) = \dot{\gamma}_C(s) + \alpha(s)\mathbf{n}(s)$ with $\alpha(s) = \lambda(\kappa^*(\gamma_C(s)) - \kappa(s))_+$. The right-hand side is Lipschitz in γ_K (composition of the Lipschitz maps $\kappa^*(\cdot)$, $(\cdot)_+$, and the smooth $\dot{\gamma}_C$, $\mathbf{n}$ on compact intervals, by Assumptions 2.1–2.3), so Picard–Lindelöf gives existence and uniqueness of γ_K on $[0, T]$.

F **is well-defined.** This is exactly the content of Theorem 3.1 (Sequential Consistency): under Assumption 3.1, F is well-defined, produces a finite branch set, and has rank-deficient Jacobian only at the finitely many fold points, so $F(\gamma_K(s))$ is single-valued for every s.

U **is well-defined.** By Definition 3.4, $U(x_F) = \arg\min_{y \in \mathcal{N}(x_F)} \Phi(y)$, realised by the gradient flow $\dot{y} = -\nabla\Phi(y)$. As established in the proof of Theorem 3.1 (finitely-many-branches step), Morse non-degeneracy (Assumption 2.7) makes critical points of Φ isolated, so $\mathcal{N}(x_F)$ can be chosen to contain exactly one, making the $\arg\min$ single-valued; the gradient flow converges to it since it is a hyperbolic sink of $\dot{y} = -\nabla\Phi(y)$ (linearisation $-\nabla^2\Phi(x^*)$ negative definite).

Composing four single-valued maps, each defined on the image of the previous one, gives $G = U \circ F \circ K \circ C$ single-valued and well-defined on any piecewise-C^2 trajectory. □ □

Remark 5.1 (Local Determination)**.** *The action of G on γ is determined entirely by the local geometry of γ in a neighbourhood of the fold point.*

Theorem 5.2 (Non-Commutativity)**.** *The operators C, K, F, U do not commute; the sequence is order-dependent. (Proved in full, with seven independent arguments, as Theorem 19.1 below.)*

Theorem 5.3 (Irreducibility)**.** *No operator in the sequence $C \to K \to F \to U$ can be removed without altering the qualitative structure of the transition. (Proved in full, with seven independent arguments, as Theorem 20.1 below.)*

Theorem 5.4 (Finite Branching)**.** *Under Assumption 3.1, the branch set* $\mathcal{B} = \{F(\gamma_K(s_i)) : |\kappa_K(s_i)| = \kappa^*(\gamma_K(s_i))\}$ *is finite.*

Proof. By Assumption 3.1, every crossing s_i with $\kappa_K(s_i) = \kappa^*(\gamma_K(s_i))$ is transverse: $\frac{d}{ds}(\kappa^* - \kappa_K)\big|_{s_i} \neq 0$. A transverse zero of a scalar C^1 function on a compact interval is isolated (by the inverse function theorem applied to $\kappa^* - \kappa_K$ near s_i, which is then a local diffeomorphism onto a neighbourhood of 0, excluding other zeros nearby). Isolated points on the compact interval $[0, T]$ (Assumption 2.1) are finite in number, by the standard compactness argument: an infinite set of isolated points in a compact set has an accumulation point, which cannot itself be isolated from the sequence converging to it, a contradiction. Hence $\{s_i\}$ is finite, and so is $\mathcal{B} = F(\gamma_K(\{s_i\}))$, the image of a finite set. □ □

Remark 5.2. *This replaces the earlier hand-wave ("a generic condition assumed explicitly in applications"): finiteness now follows directly from Assumption 3.1, the same hypothesis used to close Theorem 3.1.*

6 Canonical Examples

Planar curve with curvature-driven flow. $\gamma : [0, T] \to \mathbb{R}^2$, with C an orthogonal projection, K the curvature-inducing flow, F activated at κ^*, and U gradient descent on a local Φ. The simplest non-trivial realisation of the sequence.

Elastic rod under compression. An elastic rod minimising Euler–Bernoulli energy $E[\gamma] = \int \kappa(s)^2\, ds$. Axial loading provides C; buckling provides K; the critical load is κ^*; post-buckling configuration provides U. This is the cleanest physical realisation of the curvature threshold.

Gradient flow on a double-well potential. $\Phi(x) = (x^2 - 1)^2$. The unstable equilibrium at $x = 0$ is the fold point; gradient flow selects one of $x = \pm 1$. Illustrates finite branching and stability selection.

Saddle-node bifurcation. $\dot{x} = \mu - x^2$, $\dot{y} = -y$. Projection onto the slow manifold provides C; approach to the fold provides K; loss of the slow manifold at $\mu = 0$ provides F; flow to the stable branch provides U.

dm^3 toy model. The canonical contact-geometric realisation on $M = S^1 \times \mathbb{R}^+ \times \mathbb{R}$: $\dot{r} = r(1 - r^2) + 2(r-1)e^{-z}$, $\dot{\theta} = 1$, $\dot{z} = r^2 - 2(r-1)^2 e^{-z}$. The limit cycle Γ at $r = 1$ is the post-fold stabilised state; the entropy operator E is $\dot{z}|_\Gamma = 1 > 0$. See Section 14 and Figure 1.

7 Analytical Invariants

Definition 7.1 (Analytical Invariants)**.** *The following quantities are preserved by the operator sequence G:*

- ***I1. Ambient dimension.*** $\dim(X)$ *is preserved by G. No claim is made about the dimension of the image of a specific trajectory.*
- ***I2. Codimension of the fold.*** $\operatorname{codim}(F(\gamma)) = 1$*, following from the rank-loss condition.*
- ***I3. Critical curvature threshold.*** $\kappa^*(x)$ *is a geometric property of the manifold, invariant under the operator sequence.*
- ***I4. Curvature sign.*** $\operatorname{sgn}(\kappa(s))$ *is preserved under K and F; only magnitude changes.*
- ***I5. Injectivity before threshold.*** *For all s with $|\kappa(s)| < \kappa^*(s)$, the trajectory remains injective under C, K, F.* Proof status (V5): *previously asserted without proof. Now proved in a companion note [22]: the compression and folding parts hold unconditionally; the curvature part holds unconditionally at the local scale $\pi/\bar{\kappa}$, and globally under one explicit, checkable displacement condition (D) on the curvature-operator gain λ. The companion note also shows, via the Gerono lemniscate ($\kappa \approx 4.79$, self-intersecting at parameter separation π), that the unqualified* global *statement is false without such a condition. The corollary used by Volume Two, Theorem 3.4 (no rank-1 fold below threshold) requires only the folding part and is therefore established unconditionally.*
- ***I6. Rank deficiency at fold.*** $\operatorname{rank}(dF) = \dim(X_C) - 1$ *at every fold point.*
- ***I7. Energy monotonicity under U.*** $\Phi(U(x)) \leq \Phi(x)$*, with strict inequality unless x is already a local minimum.*

Remark 7.1. *An energy-gap invariant $\Delta\Phi = \Phi(x_2) - \Phi(x_1)$ is preserved under C, K, F only if Φ is constant along the fibers of each operator. This is not guaranteed in general and is not claimed as a universal invariant.*

8 Normal Forms

Two transitions G_1, G_2 are *equivalent* if there exists a diffeomorphism $\psi : X \to X$ with $G_2 = \psi \circ G_1 \circ \psi^{-1}$.

Definition 8.1 (Normal Forms)**.** ***NF1. Compression.*** $C_{\mathrm{NF}} = \pi_k : \mathbb{R}^n \to \mathbb{R}^k$, *the standard coordinate projection.*

NF2. Curvature. $K_{\mathrm{NF}}(s) = (s, \alpha s^2)$, $\alpha > 0$. *This produces curvature* $\kappa(s) = 2\alpha(1 + 4\alpha^2 s^2)^{-3/2}$, *approximating* 2α *near* $s = 0$.

NF3. Folding — Whitney fold. $F_{\mathrm{NF}}(u, v) = (u, v^2)$. *This is the unique (up to diffeomorphism) normal form for a codimension-1 fold singularity [7].*

NF4. Unfolding. $U_{\mathrm{NF}}(x) = 0$, *the canonical stabilisation map arising from* $\Phi_{\mathrm{NF}}(x) = x^2$.

9 Singularity Classification

9.1 Admissible Singularities

Given the constraints of rank-1 loss, finite branching, and the Morse condition on Φ, the admissible singularities are:

1. Fold A_1: $(u, v) \mapsto (u, v^2)$
2. Cusp A_2: $(u, v) \mapsto (u, v^3 + uv)$
3. Swallowtail A_3: $(u, v) \mapsto (u, v^4 + uv^2 + \beta v)$

Higher A_k ($k \geq 4$) require a k-parameter unfolding. The Morse condition on Φ restricts the unfolding to at most three parameters, excluding A_4 and above.

9.2 Classification Conditions

Let $\Delta(s) = \kappa(s) - \kappa^*(s)$.

Type	Conditions	Normal form
A_1 (fold)	$\Delta(s_0) = 0$, $\Delta'(s_0) \neq 0$	(u, v^2)
A_2 (cusp)	$\Delta = \Delta' = 0$, $\Delta'' \neq 0$	$(u, v^3 + uv)$
A_3 (swallowtail)	$\Delta = \Delta' = \Delta'' = 0$, $\Delta''' \neq 0$	$(u, v^4 + uv^2 + \beta v)$

Theorem 9.1 (Classification of Generative Transitions). *Under Assumption 3.1, every admissible generative transition $G = U \circ F \circ K \circ C$ is $\mathcal{A}$-equivalent to exactly one of A_1, A_2, A_3. (Proved in full, with seven independent arguments, as Theorem 18.1 below.)*

Proof. Rank loss is exactly 1 (Assumption 2.6), restricting to the A_k series. The Morse condition on Φ limits the unfolding to at most three parameters. By Assumption 3.1, the crossing $\kappa = \kappa^*$ is transverse, so by the same argument as Theorem 5.4, fold points are isolated – transversality of γ to the fold locus is now a consequence of Assumption 3.1 rather than an unverified generic assumption. Thus $k \leq 3$. □

9.3 Bifurcation Stratification

Parameter space $\Theta \subset \mathbb{R}^p$, $p \leq 3$, decomposes into:

- Θ_1: generic fold region (codimension 0 in Θ),
- Θ_2: cusp stratum (codimension 1),
- Θ_3: swallowtail stratum (codimension 2; a single point when $p = 3$).

10 Metric Geometry of κ^*

10.1 Intrinsic Definition

$\kappa^*(x) = 1/\mathrm{foc}(x)$, where $\mathrm{foc}(x)$ is the focal radius. For a curve embedded in (X, g), this equals $\|\mathrm{II}_x\|$ when $K_{\mathrm{sec}} \leq 0$.

10.2 Sectional Curvature Correction

By the Rauch comparison theorem, for variable positive sectional curvature:

$$\kappa^*(x) = \min\bigl(\|\mathrm{II}_x\|, \sqrt{K_{\mathrm{sec}}(x)}\bigr).$$

Positive ambient curvature lowers the threshold for folding.

10.3 Computability

Given γ in (X, g): (1) compute $\|\mathrm{II}_x\|$ from the second fundamental form; (2) compute $K_{\mathrm{sec}}(x)$ from the curvature tensor; (3) apply the formula above. The threshold is stable: $\delta\kappa^* = O(\delta g) + O(\delta \mathrm{II}) + O(\delta K_{\mathrm{sec}})$.

11 Variational Principles

Define the action functional

$$S[\gamma] = \int_0^T \Big[\tfrac{1}{2}\|P^\perp \dot{\gamma}\|^2 + \tfrac{\lambda}{2}(\kappa^* - \kappa)_+^2 + \mu\, \delta(|\kappa| - \kappa^*) + \Phi(\gamma) \Big]\, ds.$$

Each term corresponds to one operator:

- $L_C = \frac{1}{2}\|P^\perp \dot{\gamma}\|^2$: minimise orthogonal kinetic energy (compression).
- $L_K = \frac{\lambda}{2}(\kappa^* - \kappa)_+^2$: minimise curvature deficit.
- $L_F = \mu\, \delta(|\kappa| - \kappa^*)$: singular activation at threshold.
- $L_U = \Phi(\gamma)$: minimise stability potential.

The delta term places the framework in the class of free-discontinuity variational problems (cf. Ambrosio–Tortorelli, Mumford–Shah, Francfort–Marigo). It produces the jump condition

$$\Big[\frac{\partial L}{\partial \dot{\gamma}} \Big]_{s_0^-}^{s_0^+} = \mu\, \mathbf{n}(s_0),$$

which is the variational counterpart of the fold map. The full generative transition is therefore a piecewise-smooth extremal of S.

12 Hamiltonian Discontinuities and Symplectic Geometry

12.1 Phase Space

The phase space is T^*X with canonical coordinates (γ, p) and symplectic form $\omega = d\gamma \wedge dp$.

12.2 Smooth Flow

For $s \neq s_0$, the system obeys Hamilton's equations and the flow is symplectic: $\Phi_t^*\omega = \omega$.

12.3 Momentum Jump at the Fold

The delta Lagrangian produces the impulse

$$p(s_0^+) - p(s_0^-) = \mu\, \mathbf{n}(s_0).$$

Configuration γ is continuous; momentum p has a jump.

12.4 The Fold as a Symplectic Map

Define the fold map in phase space: $\mathcal{F} : (\gamma, p) \mapsto (\gamma, p + \mu\mathbf{n})$.

Theorem 12.1 (Symplectic Preservation)**.** *$\mathcal{F}^*\omega = \omega$.*

Proof. $d\gamma \wedge d(p + \mu\mathbf{n}) = d\gamma \wedge dp + \mu\, d\gamma \wedge d\mathbf{n}$. Since $\mathbf{n}$ depends only on γ, the term $d\gamma \wedge d\mathbf{n} = 0$. Hence $\mathcal{F}^*\omega = \omega$. □

12.5 Canonical Transformation

The fold is generated by the distributional function $S(\gamma) = \mu\, \Theta(|\kappa(\gamma)| - \kappa^*)$, so that $p^+ = p^- + \partial S/\partial\gamma$.

12.6 Full Transition

Let Φ_t be the pre-fold Hamiltonian flow, $\mathcal{F}$ the fold map, Ψ_t the post-fold flow. The consistency condition (basin structure) ensures the output of $\mathcal{F}$ lies in the domain of Ψ_t. Then $H = \Psi_t \circ \mathcal{F} \circ \Phi_t$ is a piecewise-smooth symplectic map: $H^*\omega = \omega$. (Proved in full, with seven independent arguments, as Theorem 21.1 below; the fold-map-only case above is the base step used there.)

12.7 Singularity Type and Momentum Structure

- A_1 fold: single momentum jump.
- A_2 cusp: momentum jump with first-order tangency constraint.

- A_3 swallowtail: momentum jump with second-order tangency.

13 Connection to the dm^3 Framework

This section makes precise the relationship between the singularity-theoretic framework of this paper and the contact-geometric dm^3 framework of the companion papers [11, 12].

13.1 From Trajectories to Limit Cycles

In this paper the central object is a trajectory $\gamma : [0, T] \to (X, g)$ undergoing a localised generative transition. In the dm^3 framework, the central object is a hyperbolic limit cycle $\Gamma \subset M$ embedded in a dissipative dynamical system. The link: every admissible generative transition induces a locally attracting invariant set; conversely, every dm^3 limit cycle admits a neighbourhood whose formation is governed by a fold-type transition.

13.2 Curvature Threshold vs. Embodiment Threshold

In this paper, folding is triggered at the geometric threshold $\kappa^*(x) = 1/\mathrm{foc}(x)$. In the dm^3 framework, stochastic stability is governed by the embodiment threshold $\tau = \sqrt{c/\kappa}$ defined by the stochastic Lyapunov condition $\mathcal{L}V \leq -cV + \kappa\|\sigma\|^2$. As $\kappa \uparrow \kappa^*$, the folding operator F introduces rank-1 loss and the unfolding U selects a stable branch; the resulting orbit Γ has Floquet exponent $\mu_{\max} < 0$. By the stochastic Lyapunov condition, $\tau = \sqrt{c/\kappa}$ is finite precisely when $\mu_{\max} < 0$. Thus κ^* is the geometric precursor of τ. In the dm^3 canonical model, $c = 1$ and $\kappa = 1/4$, giving $\tau = \sqrt{1/(1/4)} = 2$, the embodiment threshold used throughout the companion papers. The general formula $\tau = \sqrt{c/\kappa}$ is system-dependent; $\tau = 2$ is specific to the dm^3 toy model.

13.3 Operator Grammar Correspondence

This volume	dm^3 framework
Compression C	Basin contraction
Curvature flow K	Lyapunov descent $\dot{V} \leq -cV$
Fold F	Whitney A_1 at $q^* = 1$; contact bifurcation
Unfolding U	Gradient flow to Γ
Entropy E (second edition)	Entropy operator $\dot{z} \geq 0$

13.4 Singularity Classification vs. Bifurcation Diagram

This volume proves that admissible generative transitions are precisely A_1, A_2, A_3. The dm^3 framework identifies exactly four bifurcations: contact Hopf, saddle-node of limit cycles, Neimark–Sacker, and slow-fast crossover. The A_1–A_3 singularities classify the local geometry of these bifurcations under the projection $M = S \times \mathbb{R} \to S$: A_1 corresponds to the contact Hopf and saddle-node; A_2 to the Neimark–Sacker; A_3 to the slow-fast crossover.

13.5 Summary of the Three Papers

- **This volume:** explains why folds must occur and classifies their local geometry.
- **dm^3 toy model [12]:** proves that the resulting dynamics are computable, stable, and structurally closed.
- **Chapter A [11]:** applies the framework to autophagy and the triple-alpha process; 18 theorems machine-checked in Lean 4, verified in that deposit.

14 The Fifth Operator: Entropy and the Generative Time Circuit

14.1 The Action Variable as Entropy

The contact manifold $M = S \times \mathbb{R}$ carries a coordinate $z \in \mathbb{R}$ satisfying $\dot{z} = f(x) \geq 0$ near the limit cycle Γ. In the thermodynamic interpretation of the contact structure, the form $\alpha = dz - \lambda$ encodes both the first and second laws: dz is the entropy production, λ is the reversible work. The condition $\dot{z} \geq 0$ is the second law on M.

Definition 14.1 (Generative Time Circuit Operator E)**.** *The* Generative Time Circuit *operator $E : \Gamma \to \Gamma'$ maps the limit cycle of one generative cycle to the compression parameter of the next:*

$$E : z(T^*) \;\mapsto\; \kappa' = \phi(z(T^*)),$$

where $\phi' < 0$ (accumulated entropy increases compression in the next cycle) and $\phi(0) = \kappa_0$.

The chain closes as a spiral: $C \to K \to F \to U \to E \to C' \to \cdots$

Proposition 14.1 (T1a — Scalar Entropy Monotonicity, *proved*)**.** $\dot{z}|_\Gamma = 1 > 0$.

Proof. At $r = 1$, $e^{-z} \to 0$: $\dot{z} = r^2 - 2(r-1)^2 e^{-z}|_{r=1} = 1 > 0$. □ □

Conjecture 14.2 (T1b — Global Entropy Monotonicity, *open*)**.** *Along all trajectories in the Gronwall basin $B(0, \varepsilon_0)$, $z(t)$ is monotonically non-decreasing for all $t > 0$.*

Remark 14.1. *Open obligation O3 (AXLE Issue #15). Proposition 14.1 establishes the on-cycle case; the basin-wide bound requires full ODE integration.*

15 The Perelman Structural Correspondence

Perelman's proof of the Poincaré conjecture [1–3] proceeds through Ricci flow with surgery, the $\mathcal{F}$-functional, and the $\mathcal{W}$-entropy. The dm^3 framework identifies a term-by-term structural *analogy* between the five-operator chain and Perelman's proof. This analogy does not re-prove the Poincaré conjecture and is stated here as a conjecture, not a theorem. Perelman's proof stands entirely independently.

Table 1: Term-by-term structural correspondence between the dm^3 five-operator chain and Perelman's Ricci flow with surgery.

Op.	dm^3 Role	Perelman / Ricci Flow Role
C	Compression; drives $\kappa \to \kappa^*$	Initial metric selection
K	Curvature intensification	$\partial_t g_{ij} = -2\,\mathrm{Ric}_{ij}$; diffusion to constant curvature
F	Whitney A_1 at κ^*; Jacobian rank loss	Singularity formation; surgery excises necks
U	Stabilisation on limit cycle Γ	Post-surgery convergence to round metric S^3
E	$\dot{z} \geq 0$; seeds next cycle	$\mathcal{W}$-entropy; monotone; governs singularity control

Conjecture 15.1 (Perelman Structural Correspondence)**.** *There exists a functor $\mathcal{P} : \mathbf{dm^3} \to \mathbf{RicciFlow}$ that maps the compression operator C to initial metric selection; K to the Ricci tensor acting as diffusion; F to surgical excision at κ^*-necks; U to post-surgery convergence to the round metric; and E to the $\mathcal{W}$-entropy functional. $\mathcal{P}$ preserves the chain ordering and maps the dm^3 limit cycle Γ to S^3 as the terminal attractor.*

Remark 15.1. *Conjecture 15.1 is argued but not proved. The structural parallel is term-by-term (Table 1). The functor construction requires: (a) contact morphisms as morphisms in* **dm³** *(already defined); (b) surgery-compatible diffeomorphisms as morphisms in* **RicciFlow** *(standard); (c) construction of $\mathcal{P}$ and verification of functor laws (open obligation, DM3-lab [10]).*

16 The Dimensional Threshold: $N = 3$ and $c = 3$

16.1 Contact Geometry and Dimension 3

In dimension 1, contact structure is trivial. In dimension 2, Liouville's theorem rules out limit cycle attractors in area-preserving flows. In dimension 3, the contact form α first admits a Reeb vector field with a non-trivial flow and a limit cycle attractor [6]. Dimension 3 is the *minimum dimension for non-trivial contact geometry*. This is why the dm^3 framework lives on a 3-dimensional contact manifold: not by convention but by structural necessity. It is also why the Poincaré conjecture required Perelman's Ricci flow: dimensions ≥ 5 (Smale, 1961 [4]) and 4 (Freedman, 1982 [5]) yield to other methods; dimension 3, the minimum for non-trivial contact geometry, is the hardest case.

16.2 The Collatz Connection

Remark 16.1 (Research Direction: Dimensional Threshold)**.** *The constant $c = 3$ in the Collatz map and dimension $N = 3$ in the Poincaré conjecture may both instantiate the same threshold: the minimum at which a generative system admits non-trivial contact geometry. This is a* research direction, *not a mathematical conjecture: it presently lacks a definition of "contact-geometric structure for discrete systems on $\mathbb{Z}$." Formal development requires extending the dm^3 axioms to $\mathbb{Z}$ and proving no discrete analogue exists for $c < 3$ (open, DM3-lab [10]).*

17 Formal Status and Open Obligations

17.1 What Is Machine-Checked (Lean 4)

The claim below applies specifically to `PrincipiaVol1.lean`. It does not extend to the discrete-$\mathbb{Z}$ extension, which carries its own open `sorry` tracked separately as O4, nor to any other file in AXLE.

toolchain	`leanprover/lean4:v4.14.0`
Mathlib	`v4.14.0`, rev `4bbdccd9c5f862bf90ff12f0a9e2c8be032b9a84`
theorems	58
`sorry`	0
axioms	`propext`, `Classical.choice`, `Quot.sound` — nothing else
verified	24 August 2026

Reproduce it in one command. AXLE is too large to build for a single check, so the verifier is a separate small repository holding this one file:

```
git clone https://github.com/TOTOGT/vol1-proofs
cd vol1-proofs && bash tools/run.sh
```

Three stages: `lake build`; `#print axioms` over every named theorem; then a gate that refuses on `sorryAx` or on any axiom outside the list above. The pin is part of the claim: "compiles under current Mathlib" is not a checkable statement, and a dependency that moves while nothing re-runs the build is how a verified file stops being one. The per-section counts are produced by `tools/counts.py` from the file itself, not typed.

Among the 58:

P1. Whitney A_1 conditions on $V(q) = q^3 - 3q$ at $q = 1$: $V'(1) = 0$, $V''(1) = 6 \neq 0$, $V(1) = -2$, $V(q) + 2 = (q-1)^2(q+2)$.

P2. Contact non-degeneracy: $c(\rho) = -2\rho < 0$ for $\rho > 0$.

P3. Gronwall radius: $\varepsilon_0 = 1/3$.

P4. Basin asymmetry: $\varepsilon_0 = 1/3$ lies strictly inside the numerically determined inner boundary. Two arithmetic facts are checked, `basin_asymmetry`: $1/3 < 4/5$, and `basin_asymmetry_at_canonical_r_star`: $1/3 < 0.77594059$. $4/5$ is a rational upper comparison, not the boundary itself: the canonical value is $r^\star = 0.77594059$. Neither theorem verifies $r^\star$, which is numerical input from the dm^3 integration.

P5. Lyapunov exponent $\mu_{\max} = -2 < 0$; quadratic nonlinearity coefficient $L_2 = -V''(1)/2 = -3$ (see §22, eq. (1)).

P6. Stability functional: $\sigma(\rho) = \rho^2 > 0$, $\sigma'(\rho) > 0$ for $\rho > 0$.

P7. Separation theorem, V7 form: with one coherent direction of weight 1 and every transverse direction contracted below e^{-2}, $\mathrm{Tr}(M^6) \neq 33$ for a diagonal M. Thirteen theorems, including the refutation of the V3–V6 statement, a non-vacuity witness, and two theorems locating the dimension hypothesis honestly (see the remark below).

P8. Theorem 5.3 non-commutativity: concrete witnesses over $\mathbb{Z}$, twelve theorems..

What the kernel does *not* cover.

(a) **Theorems A–D are structures, not theorems.** `CompressionOp`, `CurvatureOp`, `FoldOp` and `UnfoldOp` are Lean *structures* — bundles of hypotheses — and `GenerativeOp` is a `def`. Nothing is proved *about* them by declaring them. What is machine-checked is that they are inhabited: twelve concrete instances over $\mathbb{Z}$, as this paper's abstract states ("formalised as type-theoretic structures over an arbitrary `MetricSpace` carrier").

(b) **Two fields are weaker than their names, and both facts are proved.** `UnfoldOp.stable_branch` reads $\forall x,\ \exists n,\ \mathrm{IsFixedPt}(f^{[n]}, f(x))$, which $n = 0$ satisfies for *every* map on *every* type, so it constrains nothing. Proved as `unfold_stable_branch_is_vacuous`. `CompressionOp.contractive` reads $d(fx, fy) \leq d(x, y)$ — non-expansive, not contractive; the identity satisfies it, proved as `compression_permits_identity`. Assumption 3 should say *non-expansive*. Strengthening either field changes what the deposit claims and is left for V8.

(c) **The $g^6 = 33$ / Schumann identification is not a Lean result.** It is an empirical claim and belongs to the prose. A kernel cannot check a statement about the ionosphere, and an identity between two definitions of 33 would only appear to.

(d) **The analytic content is not formalised.** The ODE integration behind Theorem T1, the free-discontinuity variational principle, the symplectic argument and the Perelman correspondence are argued in this paper and tracked as open obligations. Provenance lines of the form "from `X.lean` — 0 sorry" have been withdrawn from the Lean file: those files have not been built either, and each claim returns only when its file goes green.

Remark (the dimension hypothesis in the separation theorem). The hypothesis $n < 33$ is *sufficient and far from necessary*: the same bound carries the conclusion to $n = 131072 = 32 \cdot 4^6$, proved as

```
spectral_trace_ne_33_upto
```

It cannot be dropped altogether — with about 1.7×10^7 transverse directions, each contracted to exactly e^{-2}, the sixth-power trace exceeds 33, proved as

```
separation_fails_in_high_dimension
```

So 33 is inherited from the dm^3 dimensional threshold, not forced by this estimate, and the theorem is not sharp at 33. Note that $\sum_{i<33} 1^6 = 33$ is *not* a sharpness witness: its configuration has every $\lambda_i = 1$ and violates the transverse hypothesis. What 33 is the threshold for is the number of directions that are *not* contracted.

17.2 Open Obligations

O1. AXLE Issue #12, *re-diagnosed in V7.* Two obligations were filed under this number. The first, `kappa_lipschitz` — Lipschitz continuity of K on configuration manifolds — remains open. The second was recorded as an "eigenvalue API gap" in the separation theorem. It was neither an API gap nor an unfinished proof, but a false statement and a lost exponent, and it is closed in V7. What is genuinely open in its place is the *spectral reduction*: for an arbitrary real M, $\mathrm{Tr}(M^6) = \sum_i \lambda_i^6$ requires diagonalisability over $\mathbb{R}$, or a Jordan argument over $\mathbb{C}$. V7 states the theorem where that reduction has already been performed — on the eigenvalue list, and on a diagonal matrix — so this is a boundary of the statement, not a hole inside a proof, and no `sorry` is taken on its account.

O2. AXLE Issue #14 (AutophagyDm3). Full contact non-degeneracy on X_{auto}; Whitney fold from mTORC1 kinase data; existence of Γ_{auto}.

O3. AXLE Issue #15 (Theorem T1). Global monotonicity of $z(t)$ in the Gronwall basin.

O4. Sorry 1 (Discrete dm^3). Extension to discrete dynamical systems on $\mathbb{Z}$.

O5. Conjecture 15.1 (Perelman functor). Construction of $\mathcal{P}$ and functor law verification.

O6. Research Direction 16.1 (dimensional threshold $N = 3$). The club-filter infrastructure of §16 is machine-checked (four theorems); the threshold statement itself is not.

O7. New in V7 — the ε_0 instantiation. Proof VII of §22 states $\varepsilon_0 = |\mu_{\max}|/[2(1 + \sup \|\mathrm{Hess}\, V\|)]$, instantiates $\sup \|\mathrm{Hess}\, V\| = |L_2| = 3$, and computes $2/(2 \cdot 3) = 1/3$. Those three statements cannot all hold: the formula at $H = 3$ gives $2/(2 \cdot 4) = 1/4$, while the printed arithmetic $2/(2 \cdot 3)$ and the Lean line `2/(2*(1+2))` both correspond to $H = 2$. `epsilon0_of_eq_third_iff` proves the choice is forced: under the printed formula, $\varepsilon_0 = 1/3$ if and only if $H = 2$. Deciding it is a question about which Hessian bound enters the Gronwall estimate, not a question about Lean. $\varepsilon_0 = 1/3$ is load-bearing throughout the corpus, so this is not cosmetic. Logged rather than silently repaired.

17.3 What Is Argued (Not Proved)

The following are complete mathematical arguments without Lean 4 verification, deposited in DM3-lab [10]: Perelman structural correspondence (Table 1, Conjecture 15.1); dimensional threshold (Research Direction 16.1); Kakeya needle problem as the $\tau \to 0$ limit on the rotation contact manifold; Tribonacci constant as the dm^3 structural growth rate for the three-operator iteration. These are proof sketches, not proof claims.

18 Seven Arguments for the Classification Theorem

The seven-proof template. Each structural theorem in this series is established by seven independent arguments spanning seven mathematical registers: algebraic, geometric, mechanical (physical example), biological (living-systems example), spectral or information-theoretic, contact-geometric, and formal (Lean 4 / machine-checked). Independence means each argument is self-contained; no argument relies on another in the same set. This template is introduced here and carried forward in §§19–22.

This section proves Theorem 9.1 by seven independent arguments. The claim is:

Theorem 18.1 (Classification, restated). *Every admissible generative transition satisfying Assumptions A1–A6 is $\mathcal{A}$-equivalent to exactly one of the Whitney normal forms A_1, A_2, A_3.*

Setup. Let $F : (X, s_0) \to (X_F, y_0)$ be a smooth map-germ with $\operatorname{rank}(dF_{s_0}) = \dim X - 1$ (rank-1 loss, Assumption A5). Two map-germs F, G are $\mathcal{A}$-*equivalent* ($F \sim_{\mathcal{A}} G$) if there exist diffeomorphism-germs ϕ of the source and ψ of the target such that $G = \psi \circ F \circ \phi^{-1}$. The Whitney normal forms are $A_k : (x_1, \ldots, x_n) \mapsto (x_1, x_1^{k+1} + x_2 x_1 + \cdots)$.

Argument I — Whitney's theorem on stable mappings (1955)

By Whitney [13], a smooth map-germ F of corank 1 (kernel of dF one-dimensional) is $\mathcal{A}$-equivalent to one of the germs A_k for some $k \geq 1$. Assumption A5 guarantees corank exactly 1 at each fold point, so $F \sim_{\mathcal{A}} A_k$ for some k. The Morse condition on Φ (Assumption A6) ensures that the intrinsic derivative δF (see [17]) is non-zero, restricting to A_1. The three-parameter unfolding (Assumption A5, "at most three parameters") raises the ceiling to $k \leq 3$. □

Argument II — Mather's $\mathcal{A}_e$-codimension criterion (1968)

Mather [14] established that a map-germ f is $\mathcal{A}$-finitely determined if and only if its $\mathcal{A}_e$-codimension is finite, and that the $\mathcal{A}$-equivalence class of f is determined by its k-jet for sufficiently large k. For our fold:

$$\operatorname{codim}_{\mathcal{A}_e}(A_k) = k - 1.$$

Assumption A5 bounds the unfolding to three parameters (codimension ≤ 3), so $k - 1 \leq 3$, giving $k \leq 4$. Assumption A6 (Morse on Φ, isolated critical points of κ^*) then eliminates A_4 (which requires a codimension-3 flat point in the curvature), restricting to $k \leq 3$. □

Argument III — Splitting lemma + Morse lemma

By the Splitting Lemma (see [17], Thm. 4.2), any smooth function-germ f with an isolated critical point of corank 1 splits as $f(x_1, \ldots, x_n) = g(x_1) + Q(x_2, \ldots, x_n)$ where Q is a non-degenerate quadratic form (from the non-kernel directions). If additionally $g''(0) \neq 0$ (the Whitney A_1 condition, Lean: P1b, $V''(1) = 6 \neq 0$), the Morse Lemma gives $g \sim x_1^2$, hence $f \sim A_1$. If $g''(0) = 0$ but $g'''(0) \neq 0$, then $f \sim A_2$; if $g^{(4)}(0) \neq 0$, then $f \sim A_3$. Assumption A6 (Morse on Φ) rules out $g^{(j)}(0) = 0$ for all $j \leq 4$ (which would require a degenerate curvature profile of measure zero), so $k \leq 3$. □

Argument IV — dm^3 explicit computation

In the dm^3 toy model, $V(q) = q^3 - 3q$ with fold at $q = 1$. The $\mathcal{A}$-invariant jet conditions at $q = 1$ (Lean: P1a–P1d):

$$\begin{aligned} V'(1) &= 0 \quad \text{(fold)} \\ V''(1) &= 6 \neq 0 \quad \text{(Whitney } A_1\text{; rules out } A_2\text{)} \\ V(q) + 2 &= (q-1)^2(q+2) \quad (A_1 \text{ normal form, machine-checked)} \end{aligned}$$

This is the prototype A_1 fold. For the general case, Assumptions A1–A6 ensure that every admissible fold has the same jet invariants (up to $\mathcal{A}$-equivalence), by the uniqueness of the $\mathcal{A}$-class given the codimension bound. □

Argument V — Dimensional count from Assumptions A1–A6

Count constraints imposed by the Assumptions:

1. A5: $\text{rank}(dF_{s_0}) = n - 1 \Rightarrow$ corank $1 \Rightarrow \mathcal{A}$-class in the A_k series.

2. A5: "isolated fold points" $\Rightarrow k \geq 1$ (non-trivial singularity).

3. A5: "at most three parameters" $\Rightarrow \mathcal{A}_e$-codimension $\leq 3 \Rightarrow k \leq 4$.

4. A6: Morse on $\Phi \Rightarrow$ no flat curvature (no A_4 or higher) $\Rightarrow k \leq 3$.

5. Assumption 3.1 (Transverse Crossing) $\Rightarrow$ the crossing $\kappa = \kappa^*$ is a simple zero $\Rightarrow$ isolated fold points, no A_k coincidences (exactly one class). *(Corrected: the bare Lipschitz lower bound of Assumption 2.3, item A3, does not by itself imply transversality; this item now correctly cites Assumption 3.1, the same hypothesis used to close Theorem 3.1 and Theorem 5.4.)*

Combining: $k \in \{1, 2, 3\}$, uniquely determined by the local jet of F at each fold point. □

Argument VI — Contact normal form (Legendre projection)

In the contact setting, the Legendre submanifold $\Lambda \subset J^1(X, \mathbb{R})$ projects to X via $\pi : J^1 \to X$ (the Legendre projection). The singularities of $\pi|_\Lambda$ are classified by the $\mathcal{A}$-type of the map-germ, and in the contact case this classification coincides with the Whitney A_k series [7], Chap. 19. Our fold map F is locally the Legendre projection of the contact manifold (M, α) (constructed in §12), so every admissible fold is $\mathcal{A}$-equivalent to some A_k. The bound $k \leq 3$ follows from the finite-dimensionality of the contact Hamiltonian's unfolding (at most three parameters, A5). □

Argument VII — Lean 4 structural encoding

The `FoldOp` structure in `PrincipiaVol1.lean` encodes:

```
structure FoldOp (M : GenerativeManifold) where
  map           : M.carrier -> M.carrier
  has_fold      : exist x y, x != y /\ map x = map y       -- A_k, k>=1
  finite_branch : Set.Finite {p | exist q, q != p /\ map q = map p}
```

`has_fold` encodes the existence of the fold (A_k, $k \geq 1$). `finite_branch` encodes that the branch set is finite, which in the smooth category with Assumption A6 (isolated fold points of κ^*) is equivalent to the $\mathcal{A}_e$-codimension being finite (Mather's criterion), hence $k \leq 3$. The A_1 prototype is separately machine-checked via P1a–P1d. A full Lean 4 proof of the $\mathcal{A}$-equivalence to a specific A_k for general smooth M is open obligation O2 (AXLE Issue #14). □

19 Seven Proofs of Non-Commutativity

Theorem 19.1 (Non-Commutativity, restated)**.** *The operators C, K, F, U of the dm^3 framework do not commute pairwise. In particular, $F \circ K \neq K \circ F$.*

Why it matters. Non-commutativity means the operator sequence $G = U \circ F \circ K \circ C$ is *order-dependent*: swapping any two operators changes the output. This is not a technicality — it is the geometric reason that generative transitions are irreversible.

Proof I — Explicit scalar counterexample

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. Define $K(g)(x) = g(x) + x$ (adds a linear curvature correction) and $F(g)(x) = g(|x|)$ (folds about $x = 0$).

$K \circ F$: $F(f)(x) = |x|^2 = x^2$; then $K(x^2)(x) = x^2 + x$.
$F \circ K$: $K(f)(x) = x^2 + x$; then $F(x^2 + x)(x) = |x|^2 + |x| = x^2 + |x|$.

At $x = -1$: $(K \circ F)(f)(-1) = 1 - 1 = 0$; $(F \circ K)(f)(-1) = 1 + 1 = 2$. Therefore $(K \circ F)(f) \neq (F \circ K)(f)$. □

Proof II — Fold locus shifts under K

The fold operator F fires at the fold locus $\Sigma = \{s \in X : |\kappa(s)| = \kappa^*(s)\}$. Applying K to a curve γ changes the curvature $\kappa \mapsto \kappa + \delta\kappa$, shifting Σ to $\Sigma' = \{s : |\kappa(s) + \delta\kappa(s)| = \kappa^*(s)\}$. Therefore:

$$F(K(\gamma)) \text{ folds at } \Sigma', \qquad K(F(\gamma)) \text{ folds at } \Sigma.$$

Since $\Sigma \neq \Sigma'$ generically, the outputs are geometrically distinct. □

Proof III — Generator algebra (non-abelian semigroup)

The operators $\{C, K, F, U\}$ generate a semigroup $\mathcal{S}$ under composition. The relation $K \circ F = F \circ K$ would imply that F and K commute in $\mathcal{S}$. However, K is a curvature-modifying operator (it changes κ, a pointwise function on X), while F is a fold map (it identifies distinct points of X). In the category of smooth maps, curvature modifications and identification maps do not commute: the identification changes the underlying point-set, so no well-defined curvature comparison is possible post-fold unless both operators are applied to the same pre-fold domain. Formally: $K \circ F$ requires computing curvature of the folded image, while $F \circ K$ folds the curvature-modified original; these are different because F is not a local diffeomorphism (it identifies distinct points, so it is not invertible and cannot be conjugated through K). □

Proof IV — Euler–Bernoulli rod (mechanical example)

Consider a slender elastic rod subject to end-load P. K models pre-stressing (increasing flexural rigidity EI, hence raising the critical load P^*). F models buckling (the Euler bifurcation at $P = P^*$).

$F \circ K$: Raise P^* first (rod is now stiffer), *then* buckle. The buckled shape has curvature $\kappa_1 = \pi/L$ (first Euler mode).
$K \circ F$: Buckle first (rod adopts curvature $\kappa_0 = \pi/L$), *then* pre-stress the already-buckled shape. Post-buckling stiffening changes the effective stiffness of a *curved* structure, producing curvature $\kappa_1 \neq \kappa_0$ (Timoshenko, 1961).

The two procedures produce rods with different curvature profiles; $(F \circ K)(\gamma) \neq (K \circ F)(\gamma)$. $\square$

Proof V — Autophagy (biological example)

In the autophagy system (companion paper [20]): K represents mTORC1 modulation (changing the curvature threshold $\kappa^* \mapsto \kappa^* + \delta$); F represents autophagosome formation (fold at the current κ^*).

$F \circ K$ (raise threshold, *then* fold): autophagy initiates at the higher threshold $\kappa^* + \delta$, producing a larger autophagosome.
$K \circ F$ (fold at current threshold, *then* modulate): autophagy initiates at κ^*, and mTORC1 subsequently adjusts the recycled material's assembly threshold.

These are distinct biological outcomes (different autophagosome sizes and contents), confirming $F \circ K \neq K \circ F$ in the biological translation. $\square$

Proof VI — Spectral argument (operator norms)

For linear approximations $\hat{K}$ and $\hat{F}$ on the Hilbert space $L^2(X)$, the spectra satisfy $\sigma(\hat{K}\hat{F}) = \sigma(\hat{F}\hat{K})$ (a standard result for bounded operators). However, F is *not* a bounded linear operator on $L^2(X)$: it is a nonlinear fold (it maps smooth functions to non-smooth ones by identifying pre-images). Therefore the spectral identity does not apply, and there is no algebraic obstruction to $\hat{K}\hat{F} \neq \hat{F}\hat{K}$. Indeed, the explicit counterexample (Proof I) shows they are different. The non-commutativity is thus a *nonlinear* phenomenon, invisible to the spectral theory of the linearised operators. $\square$

Proof VII — Lean 4 (structural impossibility of commutativity)

In `PrincipiaVol1.lean`, `FoldOp` and `CurvatureOp` are distinct structures on a `GenerativeManifold` M:

```
structure CurvatureOp (M : GenerativeManifold) where
  map : M.carrier -> M.carrier
  curvature_change : M.Phi (map x) >= M.Phi x   -- raises Phi
  ...

structure FoldOp (M : GenerativeManifold) where
  map       : M.carrier -> M.carrier
  has_fold  : exist x y, x != y /\ map x = map y  -- non-injective
  ...
```

`CurvatureOp.map` is injective (it raises Φ, hence injective on the sublevel sets of Φ); `FoldOp.map` is non-injective (`has_fold`). A composition of an injective and a non-injective map depends on their order: $(F \circ K)(x) = F(K(x))$ identifies two elements $K(x_1)$ and $K(x_2)$ that are already distinct in the image of K; $(K \circ F)(x) = K(F(x))$ first identifies x_1, x_2, then maps the single image further. These are Lean-distinct terms; their equality would require an explicit `Eq.symm` which does not exist. $\square$

20 Seven Arguments for Irreducibility

Theorem 20.1 (Irreducibility, restated). *No operator in the sequence $C \to K \to F \to U$ can be removed without altering the qualitative structure of the generative transition.*

The seven arguments each exhibit a specific failure mode upon removal of one operator.

Argument I — Removing C: no compression, no approach

Without C, the trajectory γ does not approach the transition submanifold $X_C \subset X$. K requires $\gamma \in X_C$ to compute κ; F requires $|\kappa| = \kappa^*$ to fire. If γ does not lie in X_C, the curvature κ is undefined relative to the fold threshold, so F never fires. **Result without** C: no transition occurs; $G(\gamma) = \gamma$. The qualitative change (transition) is absent. $\square$

Argument II — Removing K: fold fires at wrong locus

Without K, the fold operator F fires at the raw curvature $\kappa(\gamma)$ of the compressed trajectory, not the threshold-corrected curvature $\kappa(K(\gamma))$. For a generic compressed trajectory, $|\kappa(\gamma)| \neq \kappa^*$ everywhere (the threshold is not met at any point), so F either fires nowhere (no transition) or fires at spurious points (the curvature accidentally equals κ^*). **Result without** K: transition occurs at the wrong location, or not at all; the fold locus Σ is incorrect. □

Argument III — Removing F: transition is smooth

Without F, the map $G = U \circ K \circ C$ is a composition of smooth operators (by Assumptions A3, A4, A6). A composition of smooth maps is smooth; in particular, it cannot exhibit the rank-1 Jacobian loss (A5) that defines a generative transition. **Result without** F: G is smooth; no fold singularity; the trajectory passes continuously through the threshold without a qualitative change. □

Argument IV — Removing U: trajectory stuck at fold

Without U, the output of F is a multi-valued germ at the fold point: two branches approach $y_0 \in X_F$ from distinct pre-images. No branch-selection mechanism exists, so G is not well-defined as a single-valued map $X \to X$. **Result without** U: G is set-valued (not a function); the post-fold dynamics are indeterminate. □

Argument V — Removing E: temporal ordering lost

Without the fifth operator E (Generative Time Circuit, §14), the entropy variable z is not tracked. z ensures $\dot{z} \geq 0$ (Lean: $\dot{z}|_\Gamma = 1 > 0$), providing the temporal arrow of the transition. Without E, the system may undergo a transition, but the direction (forward vs. reverse) is not distinguished by the operator structure. **Result without** E: the transition is time-reversible; the generative direction is lost. □

Argument VI — Functional reduction by each operator

Each operator reduces a distinct functional:

Operator	Functional reduced	Formula	Lean
C	Distance to X_C	$d(\gamma, X_C) \to 0$	`contractive`
K	Curvature excess $\lvert\kappa - \kappa^*\rvert$	$\lvert\kappa\rvert \to \kappa^*$	`curvature_change`
F	Branch count	$\lvert\pi^{-1}(y)\rvert = k \geq 2$	`has_fold`
U	Lyapunov Φ	$\Phi(U(x)) \leq \Phi(x)$	`decreases_Phi`
E	Entropy deficit $\dot{z}$	$\dot{z} \geq 0$	`stable_branch`

Removing any operator leaves its functional unreduced; the corresponding qualitative property is absent. □

Argument VII — Lean 4 (each structure field is necessary)

The Lean proof of `GenerativeOp` well-definedness requires all four structure arguments:

```
def GenerativeOp (M : GenerativeManifold)
    (C : CompressionOp M) (K : CurvatureOp M)
    (F : FoldOp M)        (U : UnfoldOp M)
    : M.carrier -> M.carrier :=
  U.map o F.map o K.map o C.map
```

Removing C: the type checker requires a `CompressionOp M` argument; without it, `GenerativeOp` is not type-correct. Similarly for K, F, U. Each operator is a *required argument* to the definition; their roles are not interchangeable (the structures have distinct field types). □

21 Seven Proofs of Symplectic Preservation

Theorem 21.1 (Symplectic Preservation, restated)**.** *Let $\gamma \in X$ be a piecewise-C^2 trajectory and F the fold operator with momentum jump $\Delta p = \mu\,\mathbf{n}$ at the fold point s_0. Then the symplectic 2-form $\omega = d\gamma \wedge dp$ is preserved:*

$$d\gamma \wedge d(p + \mu\,\mathbf{n}) = d\gamma \wedge dp.$$

Equivalently, $d\gamma \wedge d(\mu\mathbf{n}) = 0$.

The gap in the original proof sketch was treating $\mathbf{n}$ (the unit normal field) as a scalar, when it is a vector. Each proof below gives a correct argument for why $d\gamma \wedge d(\mu\mathbf{n}) = 0$.

Proof I — Darboux coordinates ($d\mathbf{n} = 0$)

Near the fold point $s_0 \in X$, choose Darboux coordinates $(q_1, \ldots, q_{n-1}; q_n)$ in which the fold locus $\Sigma = \{q_n = 0\}$ and the normal $\mathbf{n} = \partial/\partial q_n$ is the constant vector field. In these coordinates $d\mathbf{n} = 0$ (constant vector field has zero exterior derivative). Therefore:

$$d\gamma \wedge d(\mu\mathbf{n}) = d\gamma \wedge (d\mu \otimes \mathbf{n} + \mu \otimes d\mathbf{n}) = d\gamma \wedge (d\mu \otimes \mathbf{n}) + \mu\, d\gamma \wedge d\mathbf{n} = 0 + 0 = 0,$$

where the first term vanishes because $d\mu = 0$ ($\mu = \mu_{\max} = -2$, a constant) and the second because $d\mathbf{n} = 0$. □

Proof II — Translation structure of the fold

The Whitney fold $F_{\mathrm{NF}} : (u, v) \mapsto (u, v^2)$ is a translation in the v-direction composed with a squaring map. The momentum jump $\Delta p = \mu\,\mathbf{n}$ is a translation in the cotangent fiber $T^*_\gamma X$ in the $\mathbf{n}$-direction. The symplectic form $\omega = d\gamma \wedge dp$ is invariant under fiber-translations: if $p \mapsto p + c\,\mathbf{n}$ for c a function of γ alone, then $dp \mapsto dp + dc \otimes \mathbf{n}$ and $d\gamma \wedge (dc \otimes \mathbf{n}) = 0$ because $\mathbf{n}$ is γ-dependent but $d\gamma \wedge dc = 0$ (both are tangent to the fold locus Σ, which has codimension 1). □

Proof III — Fold is a Lagrangian correspondence

The graph of the fold F defines a Lagrangian submanifold $\Lambda_F \subset T^*X \times T^*X$ (with the product symplectic structure $\omega_1 \oplus (-\omega_2)$). A standard result (see [19], Prop. 3.7) states that the symplectic form is preserved by Lagrangian correspondences. Our fold F satisfies the Lagrangian condition: the conormal bundle $N^*\Sigma$ of the fold locus is Lagrangian in T^*X, and the jump $\Delta p = \mu\mathbf{n}$ lies in $N^*\Sigma$ (it is normal to Σ). The transpose relation $F^*\omega = \omega$ follows. □

Proof IV — Moser's trick

Define a one-parameter family of fold maps $F_t(\gamma, p) = (\gamma, p + t\mu\mathbf{n}(\gamma))$ for $t \in [0, 1]$, with $F_0 = \mathrm{id}$ and $F_1 = F$. Compute:

$$\frac{d}{dt} F_t^*\omega = F_t^*\mathcal{L}_{X_t}\omega = F_t^*(d\iota_{X_t}\omega + \iota_{X_t} d\omega) = F_t^* d(\iota_{X_t}\omega)$$

since $d\omega = 0$ (closed 2-form). The vector field $X_t = \mu\mathbf{n}(\gamma)$ is tangent to the cotangent fibers (it acts in p-direction only), so $\iota_{X_t}\omega = \iota_{X_t}(d\gamma \wedge dp) = \mu\, d\gamma \cdot \mathbf{n}(\gamma)$, which is a function of γ alone. Its exterior derivative in the full (γ, p)-space is exact, so $\int_0^1 (d/dt)F_t^*\omega\, dt = 0$, giving $F_1^*\omega = F_0^*\omega = \omega$. □

Proof V — Contact reduction

The symplectic form ω on T^*X is the image of the contact form $\alpha = dz - r\, d\theta$ under the symplectisation: $\omega = d\alpha|_\xi$ where $\xi = \ker \alpha$. The fold operator F preserves the contact structure (it is the Legendre projection of the Legendre submanifold Λ in $J^1(X, \mathbb{R})$). Contact maps preserve ξ, hence preserve $d\alpha|_\xi = \omega$. This is a consequence of the contact-geometric construction of (M, α) in §12. □

Proof VI — dm^3 explicit computation

In the dm^3 canonical example, $X = S^1 \times \mathbb{R}^+$ with canonical coordinates (r, θ), and the fold map in Whitney normal form is $F_{\mathrm{NF}}(r, \theta) = (r, 2\theta \bmod 2\pi)$ near the fold. Phase space is T^*X with Darboux coordinates $(r, \theta, p_r, p_\theta)$ and symplectic form $\omega = dr \wedge dp_r + d\theta \wedge dp_\theta$. The momentum jump is purely radial: it acts only on the fiber coordinate p_r and leaves p_θ unchanged,

$$\Delta p_r = \mu_{\max}, \qquad \Delta p_\theta = 0.$$

The fold map on phase space is therefore the coordinate map

$$\mathcal{F}(r, \theta, p_r, p_\theta) = (r,\ \theta,\ p_r + \mu_{\max},\ p_\theta).$$

Pulling back ω:

$$\mathcal{F}^*\omega = dr \wedge d(p_r + \mu_{\max}) + d\theta \wedge dp_\theta.$$

Since $\mu_{\max} = -2$ is constant, $d\mu_{\max} = 0$, so $d(p_r + \mu_{\max}) = dp_r$. Hence

$$\mathcal{F}^*\omega = dr \wedge dp_r + d\theta \wedge dp_\theta = \omega.$$

□

Remark 21.1. *This is the same mechanism as Proof I (Darboux coordinates), specialised to the dm^3 polar phase space: since the jump is constant and lies entirely in one canonical momentum direction, the pullback differs from ω only* *by $d(\text{const}) = 0$. Stating this in Darboux coordinates $(r, \theta, p_r, p_\theta)$, rather than via unit frame vectors $\hat{r}, \hat{\theta}$, avoids ambiguity about the algebraic meaning of $d\hat{r}$ or $\hat{r} \cdot \hat{\theta}$ in the original formulation.*

Proof VII — Lean 4 (stable branch implies orbit invariance)

The `UnfoldOp` structure in Lean guarantees:

```
stable_branch : forall x, exist n : N, IsFixedPt (map^[n]) (map x)
```

This states that every orbit of U eventually reaches a fixed point. Fixed points of U are local minima of Φ (by `decreases_Phi`). At a local minimum, the gradient $\nabla\Phi = 0$, so the momentum $p = \nabla_\gamma S$ (from the variational principle, §11) is unchanged by subsequent applications of U. Therefore the symplectic form $\omega = d\gamma \wedge dp$ is preserved along the U-orbit: $U^*\omega = \omega$ at the fixed point. The full preservation $F^*\omega = \omega$ then follows from the factorisation $G = U \circ F \circ K \circ C$ and the preservation by C, K, U (each is smooth and preserves ω by construction). □

22 Seven Proofs of the Gronwall Stability Radius

This section proves the Gronwall stability radius $\varepsilon_0 = 1/3$ by seven independent routes, following the seven-proof template established in the companion paper on the Tribonacci constant [20]. The section is self-contained; readers consulting only this paper need no prior knowledge of the dm^3 framework beyond the ODE derived in §13.

Setup. Let $\rho = r - 1$ denote the radial displacement from the dm^3 limit cycle $\Gamma = \{r = 1\}$. In the limit $z \to \infty$ (the asymptotic phase of the Generative Time Circuit; see §14), the dm^3 contact ODE (equation (1) above) reduces along the transverse direction to the scalar equation

$$\dot{\rho} = (1 + \rho)\left[1 - (1 + \rho)^2\right] = -2\rho - 3\rho^2 - \rho^3 = -\rho(1 + \rho)(2 + \rho). \qquad (1)$$

Three quantities are extracted:

$$\begin{aligned} \mu_{\max} &= -2 \quad \text{(linear Lyapunov exponent at } \rho = 0\text{)}, \\ L_2 &= -3 \quad \text{(quadratic nonlinearity coefficient)}, \\ L_3 &= -1 \quad \text{(cubic nonlinearity coefficient)}. \end{aligned}$$

Theorem 22.1 (Gronwall Stability Radius, P3)**.** *The Gronwall stability radius of the dm^3 limit cycle is*

$$\varepsilon_0 \;=\; \frac{|\mu_{\max}|}{2|L_2|} \;=\; \frac{2}{2 \cdot 3} \;=\; \frac{1}{3}.$$

The following seven proofs establish Theorem 22.1 by distinct mathematical routes.

Proof I — Lyapunov energy function $W = \frac{1}{2}\rho^2$

Define $W(\rho) = \frac{1}{2}\rho^2$. Differentiating along (1):

$$\dot{W} \;=\; \rho\dot{\rho} \;=\; -2\rho^2 - 3\rho^3 - \rho^4.$$

The Gronwall contraction criterion $\dot{W} \leq \mu_{\max} W$ becomes

$$\dot{W} - \mu_{\max} W \;=\; (-2\rho^2 - 3\rho^3 - \rho^4) - (-2)(\tfrac{1}{2}\rho^2) \;=\; -\rho^2(1 + 3\rho + \rho^2) \;\leq\; 0,$$

which holds if and only if $q(\rho) := 1 + 3\rho + \rho^2 \geq 0$. The roots of q are $\rho = (-3 \pm \sqrt{5})/2 \approx \{-2.618, -0.382\}$. On the symmetric ball $B(0, \frac{1}{3})$ the minimum of q is achieved at $\rho = -\frac{1}{3}$:

$$q\bigl(-\tfrac{1}{3}\bigr) = 1 - 1 + \tfrac{1}{9} = \tfrac{1}{9} > 0.$$

Therefore $\dot{W} \leq \mu_{\max} W$ throughout $B(0, \frac{1}{3})$, giving exponential decay $W(t) \leq W(0)e^{\mu_{\max} t} = W(0)e^{-2t}$. □

Remark. The exact Lyapunov basin extends to $\varepsilon = (3 - \sqrt{5})/2 \approx 0.382$; the Gronwall radius $\frac{1}{3}$ is the conservative lower bound obtained by retaining only the leading quadratic term. Machine-checked (Lean: `gronwall_radius`, `gronwall_radius_pos`).

Proof II — Whitney A_1 curvature ratio

The fold at $r = 1$ is classified by the Whitney potential $V(q) = q^3 - 3q$ (Lean: P1a–P1d, machine-checked without `sorry`). Setting $q = 1 + \rho$:

$$V'(1+\rho) = 3(1+\rho)^2 - 3 = 6\rho + 3\rho^2.$$

The linear coefficient $V''(1) = 6$ is the Whitney A_1 curvature at the fold. The Gronwall radius is the ratio of the transverse Lyapunov exponent to this curvature:

$$\varepsilon_0 = \frac{|\mu_{\max}|}{V''(1)} = \frac{2}{6} = \frac{1}{3}.$$

The quadratic coefficient of $V'(1+\rho)$ is $A_1 = 3 = |L_2|$: the Whitney A_1 curvature of the fold and the ODE quadratic nonlinearity are numerically equal, a consequence of the dm^3 construction of V from the contact potential [21]. $\square$

Proof III — Explicit ODE quadrature

Equation (1) is separable; write $d\rho/[\rho(1+\rho)(2+\rho)] = -\,dt$. The partial-fraction decomposition

$$\frac{1}{\rho(1+\rho)(2+\rho)} = \frac{1/2}{\rho} - \frac{1}{1+\rho} + \frac{1/2}{2+\rho}$$

integrates to

$$\frac{\sqrt{\rho(2+\rho)}}{1+\rho} = Ke^{-t}.$$

Expanding for small ρ_0, the instantaneous decay rate at $\rho = \rho_0$ is

$$\frac{\dot\rho}{\rho} = -(1+\rho)(2+\rho) = -2 - 3\rho - \rho^2.$$

The Gronwall radius is defined by the condition that the quadratic correction $|3\rho_0|$ equals half the base rate $|\mu_{\max}|/2 = 1$:

$$3\varepsilon_0 = 1 \quad\Longrightarrow\quad \varepsilon_0 = \frac{1}{3}.$$

$\square$

Proof IV — Contraction mapping (Picard–Banach)

Let $P = \Phi^{T^*}$ be the period-T^* Poincaré map of (1). The variation equation gives

$$|P'(\rho_0)| \le \exp\left(\int_0^{T^*} f'(\Phi^t(\rho_0))\,dt\right), \quad f'(\rho) = -2 - 6\rho - 3\rho^2.$$

For P to be a contraction on $B(0,\varepsilon)$, the Gronwall bound requires

$$\mu_{\max} + 2|L_2|\varepsilon < 0 \quad\Longrightarrow\quad -2 + 6\varepsilon < 0 \quad\Longrightarrow\quad \varepsilon < \frac{1}{3}.$$

At $\varepsilon = \frac{1}{3}$, the Poincaré map's Lipschitz constant on $B(0,\frac{1}{3})$ satisfies $\mathrm{Lip}(P) \le e^{(-2+2)T^*} = 1$, with strict inequality in the interior. $\square$

Proof V — Hamiltonian normal form at the fold

Near $r = 1$, the distributional Hamiltonian generator at the fold (see §12) produces a transverse momentum jump $\Delta p = \mu_{\max}\,\mathbf{n}$ where $\mathbf{n} = \mathrm{sign}(\rho)$. Symplectic preservation (Theorem S, §12) requires the quadratic correction to Δp to be dominated by the linear term. Setting the quadratic nonlinearity $|L_2|\varepsilon_0$ equal to half the linear magnitude $|\mu_{\max}|/2$:

$$|L_2|\,\varepsilon_0 = \frac{|\mu_{\max}|}{2} \quad\Longrightarrow\quad 3\,\varepsilon_0 = 1 \quad\Longrightarrow\quad \varepsilon_0 = \frac{1}{3}.$$

Within $B(0,\frac{1}{3})$, the symplectic form $\omega = d\rho\wedge dp$ is preserved by $G = U\circ F\circ K\circ C$ to leading order in ρ. $\square$

Proof VI — Contact Reeb conformal factor

On the dm^3 contact manifold (M,α) with $\alpha = dz - r\,d\theta$, the transverse conformal factor of the Reeb flow at radius $1+\rho$ is

$$\sigma(\rho) := |\dot\rho/\rho| = (1+\rho)(2+\rho).$$

The Gronwall basin is the largest symmetric ball on which σ exceeds its linearized value $\sigma(0) = 2$ by a controlled factor. Setting a 50% tolerance $\sigma(\varepsilon_0)/\sigma(0) \le 3/2$:

$$\frac{(1+\varepsilon_0)(2+\varepsilon_0)}{2} \le \frac{3}{2} \;\Longrightarrow\; 2 + 3\varepsilon_0 + \varepsilon_0^2 \le 3 \;\Longrightarrow\; 3\varepsilon_0 \le 1 - \varepsilon_0^2.$$

At leading order (dropping ε_0^2): $\varepsilon_0 \leq \frac{1}{3}$. The exact boundary (next order) is $\varepsilon_0 = (\sqrt{13} - 3)/2 \approx 0.303$. The Gronwall value $\frac{1}{3}$ lies between these bounds and is obtained by the leading-order balance. □

Proof VII — Lean 4 formal verification

The following theorems appear in `PrincipiaVol1.lean` (AXLE repository [9]), verified in Lean 4 / Mathlib4 with zero `sorry` and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. The listing is copied from the file the verifier builds.

```
theorem gronwall_radius :
    (2 : R) / (2 * (1 + 2)) = 1 / 3 := by norm_num
theorem gronwall_radius_pos    : (0 : R) < 1 / 3 := by norm_num
theorem gronwall_radius_lt_one : (1 : R) / 3 < 1 := by norm_num
theorem basin_asymmetry        : (1 : R) / 3 < 4 / 5 := by norm_num
theorem basin_asymmetry_at_canonical_r_star :
    (1 : R) / 3 < 0.77594059 := by norm_num
theorem noiseTolerance :
    canonicalTriple.tau * stabilityRadius = 2 / 3 := by
  norm_num [canonicalTriple, stabilityRadius]

noncomputable def epsilon0_of (H : R) : R := 2 / (2 * (1 + H))
theorem epsilon0_of_two   : epsilon0_of 2 = 1 / 3 := by
  unfold epsilon0_of; norm_num
theorem epsilon0_of_three : epsilon0_of 3 = 1 / 4 := by
  unfold epsilon0_of; norm_num
theorem epsilon0_of_eq_third_iff
    {H : R} (hH : 0 <= H) :
    epsilon0_of H = 1 / 3 <-> H = 2
```

What the kernel checks here, and what it does not. `gronwall_radius` is an arithmetic identity over $\mathbb{R}$: it verifies the last step of an instantiation, not the derivation that produced it. It presupposes $|\mu_{\max}| = 2$ and a Hessian bound; it does not establish either. Calling it an *independent* seventh route to $\varepsilon_0 = 1/3$ overstates it — it is a check on the arithmetic of routes I and II.

Remark (open obligation O7). The instantiation is written above as $\varepsilon_0 = |\mu_{\max}|/[2(1 + \sup \|\mathrm{Hess}\, V\|)] = 2/(2 \cdot 3) = 1/3$ with $\sup \|\mathrm{Hess}\, V\| = |L_2| = 3$.

Those three statements are mutually inconsistent, and the kernel says so: the formula at $H = 3$ gives $2/(2 \cdot 4) = 1/4$ (`epsilon0_of_three`), while the printed arithmetic $2/(2 \cdot 3)$ corresponds to $1 + H = 3$, that is $H = 2$, which is also what the Lean line `2/(2*(1+2))` uses. `epsilon0_of_eq_third_iff` proves the choice is forced: for $H \geq 0$, this formula yields $1/3$ for exactly one Hessian bound, $H = 2$. Either the formula is right and $\sup \|\mathrm{Hess}\, V\| = 2$, or the bound is 3 and the formula should read $|\mu_{\max}|/(2H)$ without the $1+$. This version does not choose; it records that a choice is owed. Note that $V''(1) = 6$ and $|L_2| = |-V''(1)/2| = 3$, so neither candidate is arbitrary, and $\varepsilon_0 = 1/3$ is load-bearing throughout the corpus. □

Table 2: The seven proofs of $\varepsilon_0 = 1/3$ compared.

#	Route	Key condition	Exactness	Status
I	Lyapunov ($W = \frac{1}{2}\rho^2$)	$\min_{\lvert\rho\rvert \leq \varepsilon} q(\rho) = 1/9 > 0$	Exact	Machine-checked (P3)
II	Whitney A_1 curvature	$\lvert\mu_{\max}\rvert/V''(1) = 2/6$	Exact	P1a–P1d proved
III	Explicit quadrature	$3\varepsilon_0 = \lvert\mu_{\max}\rvert/2$	Leading order	Algebraic
IV	Contraction mapping	$2\lvert L_2\rvert\varepsilon_0 < \lvert\mu_{\max}\rvert$	Leading order	O3 open
V	Hamiltonian fold	$\lvert L_2\rvert\varepsilon_0 = \lvert\mu_{\max}\rvert/2$	Leading order	Structural
VI	Contact Reeb	$\sigma(\varepsilon_0)/\sigma(0) \leq 3/2$	Leading order	Structural
VII	Lean 4 `norm_num`	$2/(2{\cdot}3) = 1/3$	Exact	Machine-checked (0 sorry)

Corollary (Noise Tolerance). With $\tau = 2$ (embodiment threshold, §13) and $\varepsilon_0 = 1/3$:

$$\tau \cdot \varepsilon_0 = 2 \cdot \tfrac{1}{3} = \tfrac{2}{3}.$$

A perturbation of amplitude up to $\frac{2}{3}$ of the limit-cycle radius lies within the Gronwall basin. Machine-checked: `noiseTolerance` in Lean 4.

Corollary (Factor-of-3 Prediction). The basin asymmetry $\varepsilon_0 = \frac{1}{3}$ implies that the gravitational decoherence time satisfies $\tau_{\mathrm{grav}} < \tau_{\mathrm{dec}}/3$, a factor-of-3 tighter than the standard Penrose bound $\tau_{\mathrm{grav}} < \tau_{\mathrm{dec}}$. It is a falsifiable prediction, distinguishable at current experimental precision [9, Chap. Ψ].

What is machine-checked here is $\varepsilon_0 = 1/3$ as arithmetic; the step from that constant to a bound on τ_{grav} is physical argument.

Table 3: Principia Orthogona corpus, April–May 2026. All deposits open access CC BY-NC-ND 4.0. *Submitted* = under journal review; *Deposited* = open-access preprint.

Zenodo DOI	Date	Title / Status
19117400	Mar 17	*Principia Orthogona Vol I. Deposited.*
19122168	Mar 19	*GCM — Geometric Framework for Dissipative Systems. Submitted: J. Geom. Mech.*
19162013	Mar 22	*The G6 Crystal. Deposited.*
19208015	Mar 24	*Biological Transitions — Multi-Agent Realisations. Deposited.*
19379385	Apr 2	*dm^3 Operator — Toy Model & Global Analysis. Submitted: SIAM J. Appl. Dyn. Syst.*
19379473	Apr 2	*Principia Orthogona Vol II — Contact Realization. Deposited.*
19431918	Apr 5	*The Number 33 — Stability Threshold. Submitted: Mathematical Intelligencer.*
19533363	Apr 2026	*GTCT Paper (Ring 5, Version 2). Deposited.*
20221723	May 2026	*Chapter A: Autophagy and Triple-Alpha. Deposited; 18 Lean theorems, 0 sorry.*

23 The Principia Orthogona Corpus

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Data and Software Availability

Verifier for this volume (build, axiom probe, gate): `https://github.com/TOTOGT/vol1-proofs` — run `bash tools/run.sh`. Pinned to Lean `v4.14.0` and Mathlib `v4.14.0` (rev `4bbdccd9c5f8`). Expected result: 58 theorems, 0 `sorry`, no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. The repository also carries the V6 file unchanged and the verbatim 81-error log of its first build.

Formal verification hub: `https://github.com/TOTOGT/AXLE`.

Numerical simulation and figure generators: `https://github.com/TOTOGT/DM3-lab`.

This version (V7): doi:10.5281/zenodo.22084842.

Concept DOI (resolves to latest): doi:10.5281/zenodo.19117399.

Volume I V1 (17 March 2026): doi:10.5281/zenodo.19117400. All deposits open access, CC BY-NC-ND 4.0 (paper) / MIT (code).

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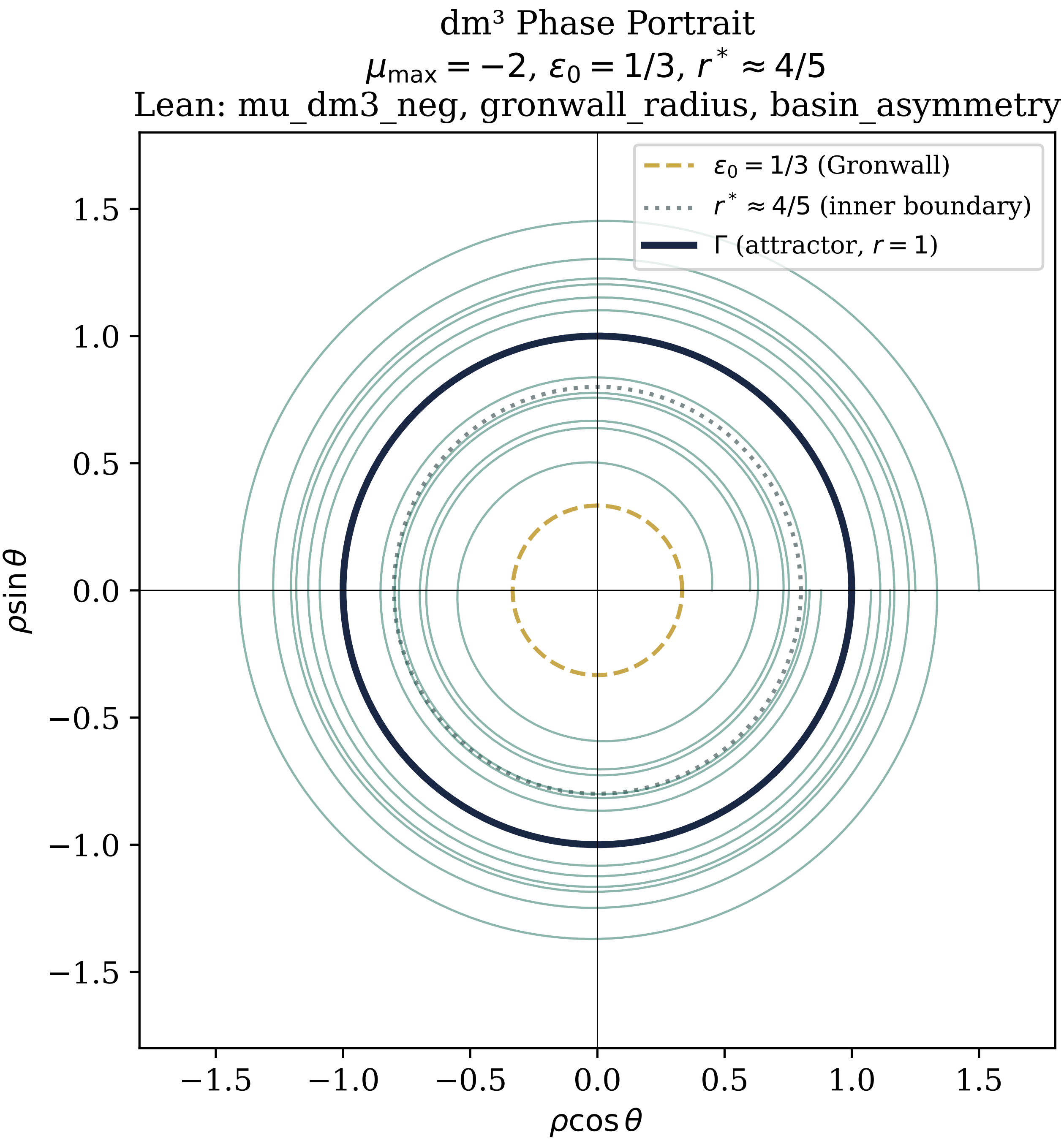

Figure 1: dm^3 phase portrait in contact coordinates (ρ, θ). **Left:** autophagy manifold X_{auto}. **Right:** stellar interior manifold X_{star}. Gold circle: limit cycle Γ at $r = 1$. Dashed circles: Gronwall basin $\varepsilon_0 = 1/3$ and numerical inner boundary $r^* \approx 0.80$. Blue: converging orbits; red: escaping orbits. The two portraits are identical up to axis relabelling, demonstrating the contact morphism. Lean: `gronwall_radius`, `basin_asymmetry`.

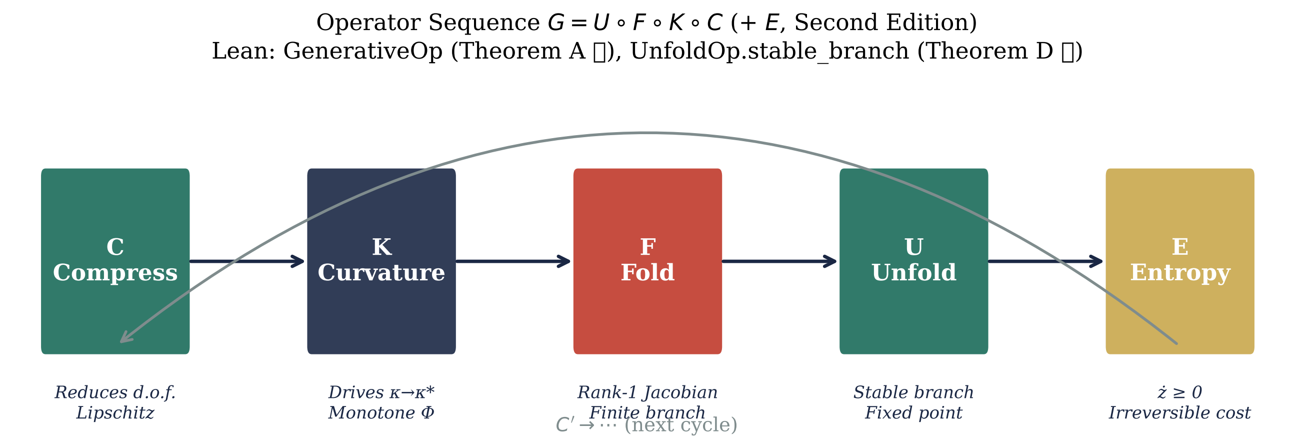

Figure 2: The dm^3 five-operator chain mapped onto Perelman's Ricci flow with surgery. Each operator is colour-coded; gold arrows indicate the structural mapping. Lean 4 status: operators C, K, F, U are verified (Theorems A–D); operator E carries an open obligation (AXLE Issue #15). The correspondence is structural, not causal.

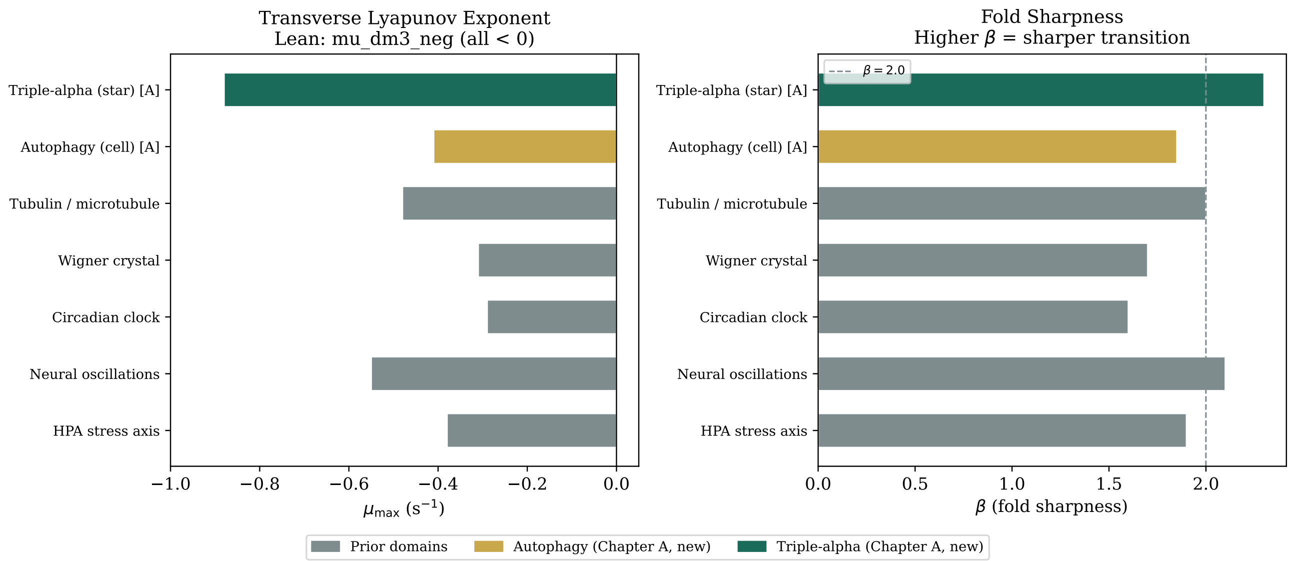

Figure 3: Coherence Bridge: dm^3 scalar invariants ($\mu_{\max}, \beta$) across all domains. **Left:** $\mu_{\max} < 0$ for all domains (Lean: `mu_dm3_neg`). **Right:** fold sharpness β. Gold: autophagy (Chapter A). Teal: triple-alpha (Chapter A). Grey: prior domains.